

Beckham Moreland

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Education

San Diego State University, College of Engineering
B.S. in Aerospace Engineering

Expected May 2029
Cumulative GPA: 3.72/4.0

Experience

UAS Engineering Intern, NASA Ames (AMA) – Moffett Field, CA Jun 2026 – Aug 2026

- Integrated and revalidated avionics across 3 evolving PX4 UAS builds, debugging motor/ESC, telemetry-link, battery-sensing, vibration, magnetometer, and altitude-drift issues through functional tests and ULog analysis
- Planned and executed 10+ hover, handling, CG, landing, and drop tests plus 3 aerial releases from 30+ ft, using test cards, go/no-go criteria, live telemetry, and post-flight logs to drive configuration changes
- Built a Python/PyULog flight-test application with data-quality checks, database-backed log management, interactive state/actuator plots, and event detection to compare behavior across 3 prototype iterations
- Developed and validated a PX4 SITL recovery controller using Python, Gazebo, and MAVLink, adding mode-transition checks and failure handling while tuning detection logic from real Black Hawk IMU data
- Led a 15+ run UAS stability campaign over 18 wind-on hours, varying speed, CG, yaw, and airframe configuration from 10–85 ft/s to expose stability limitations before flight integration
- Supported NASA flight operations with Skydio X2 and Freefly Alta X aircraft, performing aircraft setup, battery/camera preparation, maintenance documentation, BVLOS/VO support, and telemetry-readiness checks

Test and Visualization Engineering Intern, MeshMap Labs Inc. – San Francisco, CA May 2025 – Sep 2025

- Led UAV/RTK field testing across 4 sites, varying altitude, image overlap, and capture parameters to verify collection performance and establish repeatable operating procedures under changing conditions
- Diagnosed coverage and data-quality failures through post-flight analysis, planned corrective missions, and verified revised acquisition workflows in georeferenced UAV/LiDAR outputs

Full Stack Developer Intern, Axyz Technologies – Oakland, CA Jun 2023 – Jul 2025

- Developed and tested 12+ Python/Django and JavaScript web features across database-backed customer workflows, reproducing defects, validating fixes, and iterating from stakeholder requirements

Projects

Fault-Tolerant Navigation Bench, Personal Project In Progress

- Developing a GNSS/IMU validation bench with Python fault injection for GPS loss, sensor bias/noise, frozen telemetry, and signal dropouts to characterize detection and recovery behavior

Autonomous UAS Systems Research, SDSU Undergraduate Research Mar 2025 – Present

- Integrating sensors, telemetry links, and ground-station controls on an experimental UAV while documenting subsystem interfaces and validating communications ahead of hardware and flight testing

Skills

Flight Test & UAS: PX4, QGroundControl, UAS flight testing, test plans/cards, readiness/go-no-go checks, functional testing, system revalidation, flight-log analysis, telemetry/wireless testing, RTK/GNSS, BVLOS/VO

Avionics & Integration: Flight controllers, ESC/DShot, BLDC propulsion, IMU/magnetometer/GPS, battery/power sensing, telemetry radios/antennas, sensor/actuator interfaces, soldering/wiring

Software & Test: Python, Linux (WSL), PyULog, MAVLink, MAVProxy, Gazebo, pandas, Matplotlib, Django, JavaScript, Git/GitHub, test automation, web-application testing